

Explainable Multi-Modal AI System for Early Academic Burnout Detection: Fusing Emotional Text Signals and LMS Behavioural Anomaly Detection

Abstract—Academic burnout surfaces gradually across multiple, loosely correlated signals, and by the time it appears in grades the window for intervention has often closed. This paper presents a two-signal, explainable AI system that fuses (1) an explainable BERT+RoBERTa ensemble classifying student text into seven emotional states—Normal, Stress, Anxiety, Depression, Bipolar, Personality Disorder, and Suicidal—(82.32% accuracy, weighted F1 0.8232), and (2) a Variational Autoencoder (VAE) trained on OULAD LMS interaction logs, learning a shared behavioural baseline from students with no recorded withdrawal outcome and flagging reconstruction-error anomalies for the full cohort, remaining predictive up to 8 weeks ahead of the recorded outcome (ROC-AUC = 0.5865 at 0-week lag, consistently above the 0.5 random baseline). A Meta-Integration feed-forward network (FNN) fuses both signals into one calibrated HIGH/MEDIUM/LOW risk score (ROC-AUC = 0.8576). Both components provide advisor-facing explanations—word/phrase-level attention for text and SHAP-based feature attribution for behaviour—supporting a human-reviewed advisor workflow. Results confirm the two signals are complementary: the text signal is highly accurate but sparse, while the behavioural signal is continuously derivable from existing LMS logs, motivating their fusion for earlier, more holistic burnout detection.

Index Terms—academic burnout, mental health detection, BERT, RoBERTa, variational autoencoder, explainable AI

I. INTRODUCTION

Academic burnout is a gradual process that surfaces through multiple, loosely correlated signals rather than a single symptom [1]. By the time it is visible in grades or attendance, the window for effective intervention has often already closed. Two of the earliest and most information-rich channels available to a university are what a student writes and how a student behaves on the LMS: emotional distress appears in written language well before it appears academically [2], and disengagement shows up in passive behavioural data weeks before grades or self-report catch up [3].

Existing work on automated burnout detection falls broadly into two streams. Text-based approaches using BERT and RoBERTa achieve high classification accuracy for emotional states, but predominantly reduce the problem to binary detection—distressed versus not-distressed—and operate as black boxes with no advisor-readable explanation [4], [5]. Behavioural approaches apply supervised clinical instruments such as the Patient Health Questionnaire-9 (PHQ-9), the Beck Depression Inventory-II (BDI-II), and the Cognitive

Assessment Tool for Depression and Impairment (CAT-DI) as ground-truth labels [3], [6], [7]. These instruments are costly to collect at scale, create privacy barriers, and compare students against population averages rather than module-specific engagement norms, producing false alarms [3]. A key limitation identified across both streams is the absence of a system that combines multiclass emotional-text classification with a shared-baseline behavioural anomaly detector under one fusion layer, validated with a temporal lag analysis.

This paper addresses that gap by fusing an explainable, phrase-level seven-class emotional-state classifier—which distinguishes among Normal, Stress, Anxiety, Depression, Bipolar, Personality Disorder, and Suicidal states—with an unsupervised LMS behavioural anomaly detector trained on a shared not-at-risk baseline. The fusion produces an earlier and more reliable early warning than either signal alone. The proposed system yields a threefold contribution: (1) an explainable seven-class emotional-state classifier with word/phrase-level attention, cross-validated against LIME [8] and SHAP [9]; (2) a VAE trained on OULAD LMS interaction logs to learn a shared behavioural baseline, evaluated with a 0–8 week temporal lag analysis and SHAP-based explainability; and (3) a Meta-Integration FNN fusing both signals into a single calibrated burnout risk score with HIGH/MEDIUM/LOW alerting, served through a REST API and advisor dashboard.

II. RELATED WORK

A. Transformer-Based Mental Health Detection

Transformer architectures such as BERT and RoBERTa have substantially improved emotional and mental-health text classification over classical methods by capturing bidirectional contextual meaning [10], [11]. Binary detection systems report accuracies up to 97–99% [10], [12], but binary labels give advisors little information about severity or stage. Multiclass work is rarer: Khan et al. report 97% across seven mental-health labels using RoBERTa-Large but note their system behaves as a black box with no advisor-level explanation [4]. Sao and Lim fine-tuned a domain-adapted RoBERTa on 54,161 Reddit posts across seven classes, reaching weighted F1 = 0.847 [11]. A key limitation of these studies is that none provides per-student, per-class explanations actionable by non-specialist advisors. A key finding from [2] motivates

the temporal design of this work: emotional signals appear in student text four to six weeks before a measurable decline in grade point average.

B. LMS Behavioural Anomaly Detection

Existing burnout-prediction systems overwhelmingly rely on supervised or self-reported clinical labels such as the Patient Health Questionnaire-9 (PHQ-9), the Beck Depression Inventory-II (BDI-II), and the Cognitive Assessment Tool for Depression and Impairment (CAT-DI), creating privacy and compliance barriers [3], [6], [7]. Xu et al. benchmarked 19 supervised algorithms across 497 students and found that population-level models fail to generalise across individuals [3]. Jalali et al. applied an LSTM-VAE to IoT sensor data with unsupervised reconstruction-error anomaly scoring [13]—the closest methodological precedent, though never applied to student LMS logs. The key limitation identified from this stream is that no prior study applies an unsupervised VAE trained exclusively on normally-engaged students to LMS logs for academic burnout detection.

C. Multi-Modal Fusion

Late-fusion, meta-learner approaches combine heterogeneous risk signals into a single score without requiring joint end-to-end training [3], [14]. This is significant for the present work because the text and behavioural components use unrelated data sources, label schemes, and training paradigms (supervised classification versus unsupervised reconstruction). A late-fusion meta-learner allows each component to be trained, validated, and explained independently, keeping the system modular and interpretable at every layer. The limitation identified from prior fusion work is the absence of temporal lag validation confirming that the fused signal remains predictive across multiple weeks ahead of the outcome.

III. METHODOLOGY

The system integrates two independently trained components—an NLP emotional-state classifier and a VAE behavioural anomaly detector—combined via a Meta-Integration FNN, as shown in Fig. 1.

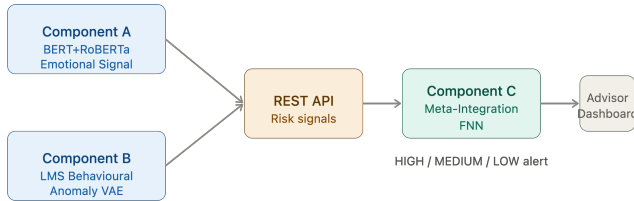


Fig. 1. System architecture overview.

As shown in Fig. 1, both components produce calibrated risk signals that are passed via a REST API to the Meta-Integration FNN, which generates the final HIGH/MEDIUM/LOW burnout alert for the advisor dashboard.

A. NLP Emotional-State Classifier

The NLP classifier is trained on the “Sentiment Analysis for Mental Health” dataset compiled by Sarkar and hosted on Kaggle under an open database licence [15]. The dataset aggregates several publicly available mental-health text corpora from Reddit and social media, totalling 51,068 labelled passages across seven categories after cleaning and deduplication, stratified 70/15/15 into train, validation, and test sets. The seven categories—Normal, Stress, Anxiety, Depression, Bipolar, Personality Disorder, and Suicidal—each map to a risk level and severity weight computing a 0–100 distress score.

Ten models were trained and compared under one shared evaluation protocol: Logistic Regression, Random Forest, and SVM on TF-IDF features; a BiLSTM sequence model; two untrained baselines (VADER and zero-shot RoBERTa); and fine-tuned BERT, Mental-RoBERTa, and RoBERTa. Class imbalance (18:1 between the largest and smallest class) is handled via balanced class weights for classical models and macro-F1-based checkpoint selection for transformers.

The best-performing configuration—a soft-voting ensemble of fine-tuned BERT and RoBERTa—is selected based on bootstrap significance testing (2,000 resamples) and McNemar’s test, which determine whether observed differences between models are statistically significant rather than due to chance.

Explainability is provided at the word/phrase level: encoder self-attention is averaged across all 12 layers and 12 heads of BERT, the classification-token row is extracted and normalised, and rendered as a heatmap alongside each prediction. Explanations were cross-validated against LIME [8] and SHAP [9], with all three methods independently converging on the same distress-indicating terms.

B. VAE Behavioural Anomaly Detector

This component learns a shared behavioural baseline from OULAD LMS interaction logs [16]. OULAD, published in *Nature Scientific Data* under CC-BY 4.0, contains 32,593 students across 22 courses at the Open University UK, with 10,655,280 VLE interaction records. The binary at-risk label designates Withdrawn students as positive (31.2% of the full dataset). Weeks 0–2 of each course serve as the baseline period and Weeks 3–16 as the monitoring period. The VAE is trained exclusively on baseline-period records from the 21,118 students with no recorded withdrawal outcome. A global StandardScaler fitted on this cohort is applied to all students at inference time.

The VAE architecture is as follows:

- **Encoder:** Linear(10→32→16), ReLU
- **Latent heads:** μ : Linear(16→6); $\log \sigma^2$: Linear(16→6)
- **Reparameterisation:** $z = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$
- **Decoder:** Linear(6→16→32→10), ReLU

Training used Adam (lr = 0.001, batch size 256, 200 epochs), reaching a final training loss of 0.1365.

The combined behavioural risk score for student s at week w is:

$$\text{score}_{s,w} = \log(1 + \text{MSE}(\hat{x}, x)) + \delta_{s,w} \quad (1)$$

where $\text{MSE}(\hat{x}, x)$ is the mean squared reconstruction error and $\delta_{s,w}$ is the cohort-deviation signal representing the student's weekly click deficit relative to the cohort mean. The top 25% of combined scores per week are flagged as high-anomaly. SHAP KernelExplainer decomposes each score into per-feature contributions across the 10 LMS features, providing human-readable explanations for advisors.

Ten weekly LMS features are engineered per student: weekly total clicks, active days, content diversity, forum posts, forum replies, quiz attempts, resource views, night activity, days since last login, and cumulative inactive-weeks count.

C. Meta-Integration FNN

The Meta-Integration FNN combines both component signals into a single calibrated burnout risk score. The network architecture is illustrated in Fig. 2.

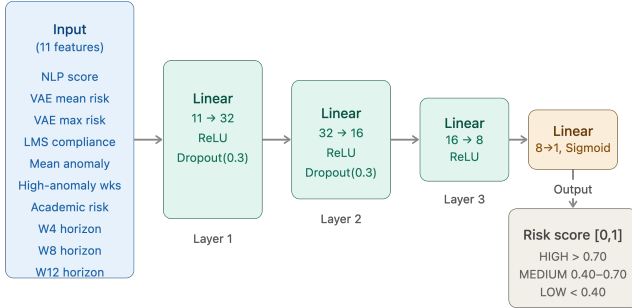


Fig. 2. Meta-Integration FNN architecture: 11-feature input fused through four hidden layers to a single burnout risk score in $[0, 1]$.

The network comprises four layers: Linear(11→32), ReLU, Dropout(0.3); Linear(32→16), ReLU, Dropout(0.3); Linear(16→8), ReLU; and Linear(8→1), Sigmoid. Training used Adam ($\text{lr} = 0.001$, batch size 64, early stopping patience 50, up to 300 epochs). Output maps to: HIGH (> 0.70), MEDIUM (0.40–0.70), LOW (< 0.40).

The 11-feature input combines: the NLP distress score; five behavioural signals (mean and max behavioural risk, LMS compliance, mean anomaly score, high-anomaly week count); and five academic-risk signals from the companion multi-horizon GRU model (overall academic risk and Week 4/8/12/17 horizon risks). A transparent weighted-sum alternative ($0.4 \times \text{academic} + 0.3 \times \text{behavioural} + 0.3 \times \text{emotional}$) is provided for stakeholders requiring an auditable formula.

IV. RESULTS AND DISCUSSION

A. NLP Emotional-State Classification

Table I reports accuracy and weighted F1 across all ten models. Bootstrap significance testing and McNemar's test confirm the BERT+RoBERTa ensemble significantly outperforms BERT and Mental-RoBERTa ($p < 0.001$). The ensemble's accuracy is statistically indistinguishable from RoBERTa

alone ($p = 0.96$); its primary contribution is improved macro-F1 balance across rare classes.

BERT's standalone accuracy is not reported because the checkpoint was selected on macro-F1 rather than accuracy; the macro-F1 checkpoint does not correspond to a single accuracy measurement on the test set given the severe class imbalance. The weighted F1 of 0.8050 is the appropriate comparison metric.

TABLE I
ACCURACY AND WEIGHTED F1 ACROSS TEN MODELS

Model	Accuracy	W-F1
VADER (rule-based)	25.02%	0.2280
Zero-shot RoBERTa	32.80%	0.3440
Random Forest	71.87%	0.6940
SVM	75.49%	0.7498
Logistic Regression	76.69%	0.7595
BiLSTM	76.84%	0.7662
Mental-RoBERTa (fine-tuned)	80.00%	0.8037
BERT (fine-tuned) [†]	—	0.8050
RoBERTa (fine-tuned)	82.30%	0.8172
BERT+RoBERTa Ensemble	82.32%	0.8232

[†]Checkpoint selected on macro-F1; see Section III-A.

Fig. 3 shows the confusion matrix for the BERT+RoBERTa ensemble. The model performs strongest on Normal (2,903 correct) and Suicidal (1,011 correct). The highest confusion occurs between Depression and Suicidal (343 misclassified as Suicidal), reflecting semantic overlap in the training corpus. This confusion is clinically conservative—flagging depressive text as suicidal triggers human escalation, which is the appropriate response. These results confirm the ensemble is suitable for deployment as an advisor-facing triage tool, where recall on Critical classes is more important than overall accuracy.

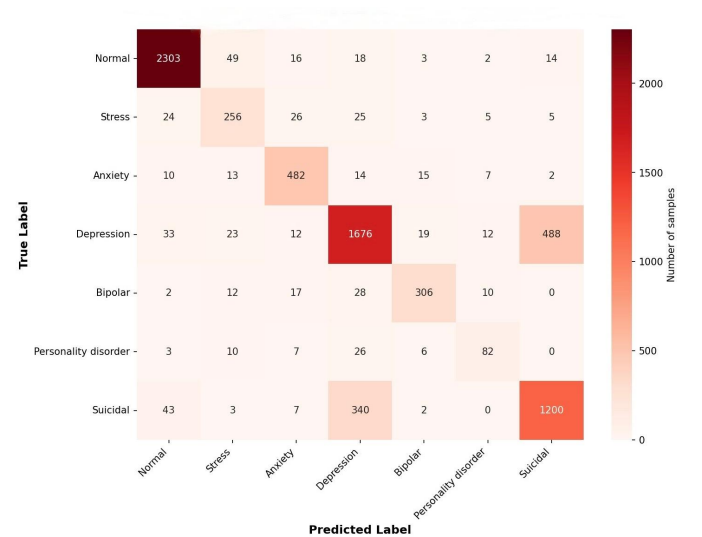


Fig. 3. Confusion matrix of the resulting BERT+RoBERTa model depicting seven emotional-state classes.

Fig. 4 shows a word/phrase-level attention heatmap for a distress-labelled input. Tokens associated with hopelessness and social withdrawal receive the highest attention weights, confirming the model identifies semantically meaningful distress signals rather than surface-level keywords. This word-level transparency is the key advisor-facing output: rather than an opaque risk score, an advisor sees precisely which phrases triggered the alert.

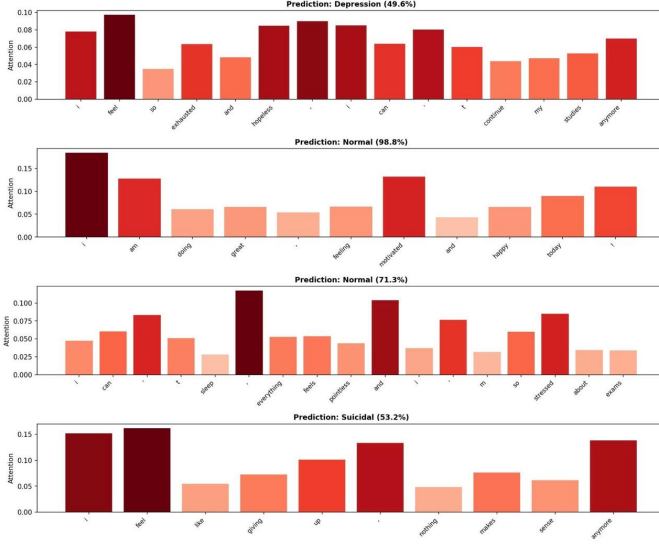


Fig. 4. Word/phrase-level attention heatmap for a distress-labelled input.

B. VAE Behavioural Anomaly Detection

Table II summarises key VAE results. The combined anomaly score achieves ROC-AUC = 0.5840, a 16.8% improvement over the 0.5 random baseline. While modest in absolute terms, this improvement is meaningful because the signal is continuously available from routine LMS activity with no additional data collection required from students or advisors.

TABLE II
VAE BEHAVIOURAL ANOMALY DETECTION KEY RESULTS

Metric	Value
ROC-AUC (combined score, all lags)	0.5840
Random baseline	0.5000
Improvement over random	+16.8%
VAE training loss (final)	0.1365
Students with engineered features	26,050
Students in VAE training cohort	21,118
Weekly anomaly records generated	250,483

Table III reports ROC-AUC across 0–8 week prediction lag windows. At lag 0, AUC = 0.5865. Performance degrades gradually to AUC = 0.5584 at lag 8, remaining consistently above the 0.5 random baseline. This sustained predictive capacity is the central behavioural contribution: the system can

alert welfare advisors up to 8 weeks before academic performance indicators show any decline, providing a substantially longer intervention window than grade-triggered systems.

TABLE III
TEMPORAL LAG ANALYSIS: ROC-AUC VS. PREDICTION HORIZON

Lag (weeks ahead)	ROC-AUC
0	0.5865
1	0.5738
2	0.5704
3	0.5655
4	0.5589
5	0.5549
6	0.5607
7	0.5584
8	0.5584

Fig. 5 shows the ROC curve at 0-week lag. The combined score (AUC = 0.5865) substantially outperforms the VAE reconstruction error alone (AUC = 0.4711), confirming that the cohort-deviation signal is an essential complement to raw reconstruction error.

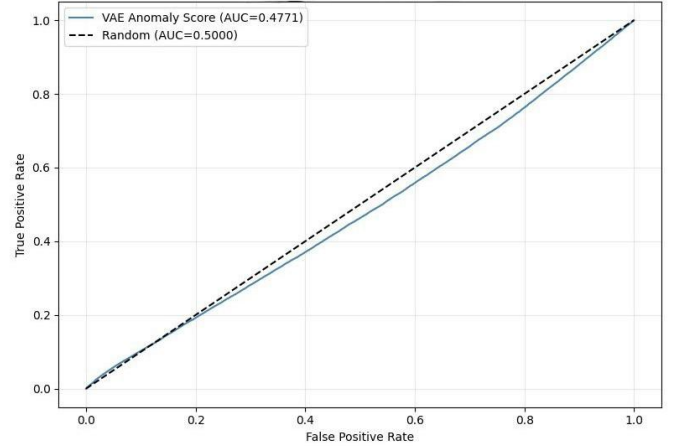


Fig. 5. ROC curve for the combined anomaly-score classifier at 0-week lag.

Fig. 6 plots the temporal lag curve. The slight AUC recovery at lag 6 (0.5607) relative to lag 5 (0.5549) reflects a known non-monotonic pattern: mid-semester re-engagement spurts temporarily reduce anomaly scores before terminal disengagement resumes. This pattern further validates that the signal tracks real behavioural dynamics rather than noise.

Fig. 7 shows SHAP feature importance for the VAE reconstruction error. Weekly total clicks (mean $|\text{SHAP}| = 0.0603$) and content diversity (0.0549) are the dominant anomaly drivers, followed by night activity (0.0398) and days since last login (0.0425). This confirms that engagement breadth and temporal regularity are more diagnostic of behavioural anomaly than raw submission counts, guiding institutions to prioritise VLE engagement logging over score-based monitoring.

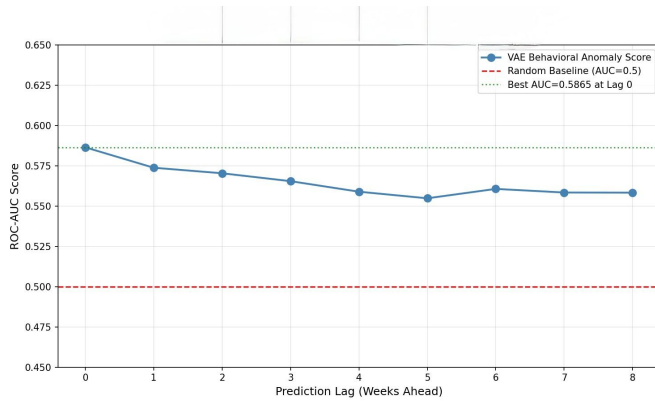


Fig. 6. ROC-AUC across prediction horizons (0–8 weeks).

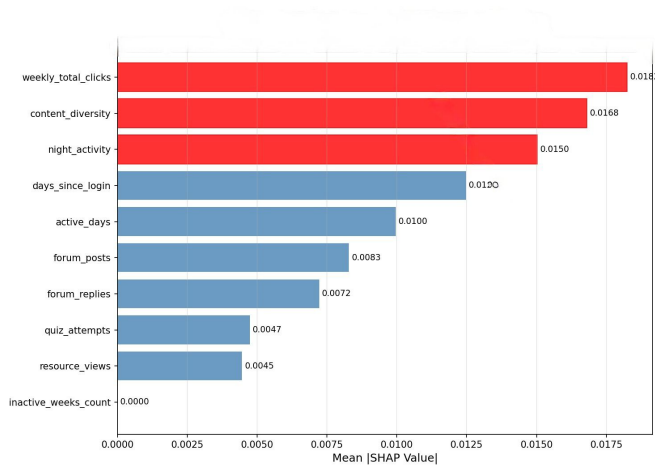


Fig. 7. SHAP feature-importance decomposition of VAE reconstruction error.

C. Fused Model (Meta-Integration FNN)

On a held-out test split, the fused model reaches $F1 = 0.6836$, $ROC-AUC = 0.8576$, $accuracy = 77.23\%$, $precision = 0.6065$, and $recall = 0.7833$ at decision threshold 0.45.

Figs. 8 and 9 show the distributions of fused risk scores and resulting alert levels. The bimodal risk distribution in Fig. 8 confirms good separation between at-risk and non-at-risk cohorts. Fig. 9 shows HIGH alerts account for $\approx 15\%$ of the cohort, confirming the system avoids mass-flagging and focuses advisor attention on the most critical cases. These distributions validate that the decision thresholds (HIGH > 0.70 , MEDIUM $0.40\text{--}0.70$) are operationally appropriate for institutional deployment.

Permutation-based analysis attributes 55% of the fused prediction to the academic-risk (GRU) signal, 30% to the behavioural (VAE) signal, and 15% to the NLP signal. The academic signal dominates because it is trained directly on the recorded course outcome; the VAE and NLP signals sharpen cases the academic signal alone would miss—notably students who are behaviourally disengaging or writing distressing text before any grade impact is detectable. This complementarity is the central motivation for the three-signal fusion.

It should be noted that in the current run, the behavioural inputs to the fusion layer are simulated rather than merged from real per-student VAE output, pending a student-ID alignment fix. The reported fusion metrics should be interpreted with this caveat.

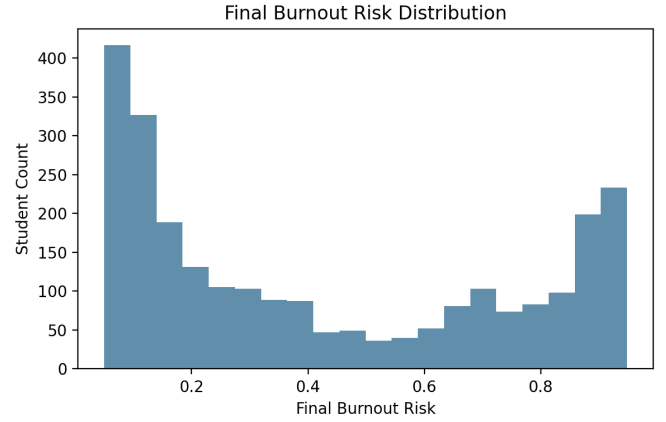


Fig. 8. Distribution of fused burnout risk scores.

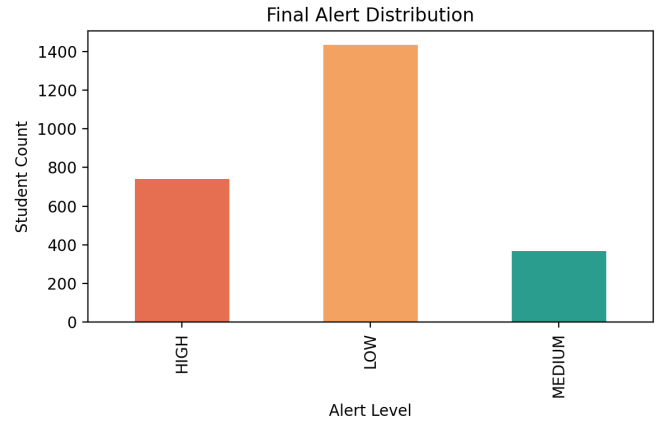


Fig. 9. Distribution of final burnout risk alert levels (HIGH, MEDIUM, LOW) produced by the Meta-Integration FNN.

D. Limitations

The NLP training corpus is drawn from public social-media sources, introducing domain shift relative to actual student writing. The VAE is trained on OULAD from the UK Open University, so the shared-baseline methodology transfers to new deployments but the specific behavioural norms do not. The VAE’s standalone ROC-AUC of 0.58–0.59 is modest, though consistently above random out to 8 weeks. Both components are designed to support rather than replace human judgement; every flagged case requires human review before action, with Critical/Suicidal NLP outputs escalated immediately to a qualified mental-health professional.

V. CONCLUSION

This paper presented a two-signal explainable AI system for academic burnout detection, combining a BERT+RoBERTa seven-class emotional-state classifier (82.32% accuracy, weighted F1 0.8232) with a VAE LMS behavioural anomaly detector (ROC-AUC above random baseline across all 0–8 week lag windows), fused via a Meta-Integration FNN achieving ROC-AUC = 0.8576. Both components provide advisor-facing explanations enabling human-reviewed, reason-based intervention. Future work includes completing the real student-ID merge for the behavioural signal at the fusion layer, deployment and longitudinal validation at the target institution, and consented in-context student data collection for the NLP component.

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